



Figure 3.4: S_1 and S_2

Then

$$S_1 \cap S_2 = \{(0, y) : 0 \leq y \leq 1\}$$

has zero content.

Hence, by using corollary implies that if f is integrable on S_1 and S_2 , then f is also integrable over

$$S = S_1 \cup S_2 = \{(x, y) : -1 \leq x \leq 1, 0 \leq y \leq 1 + |x|\}$$

and

$$\int_{S_1 \cup S_2} f(\mathbf{X}) d\mathbf{X} = \int_{S_1} f(\mathbf{X}) d\mathbf{X} + \int_{S_2} f(\mathbf{X}) d\mathbf{X}.$$

3.13 Iterated Integrals

Let us first assume that f is continuous on $R = [a, b] \times [c, d]$.

Then, for each y in $[c, d]$, $f(x, y)$ is continuous with respect to x on $[a, b]$, so the integral

$$F(y) = \int_a^b f(x, y) dx$$

exists.

Moreover, the uniform continuity of f on R implies that F is continuous and therefore integrable on $[c, d]$.

We say that

$$I_1 = \int_c^d F(y) dy = \int_c^d \left(\int_a^b f(x, y) dx \right) dy$$

is an *iterated integral* of f over R .

Iterated integrals: We will usually write it as

$$I_1 = \int_c^d dy \int_a^b f(x, y) dx.$$

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Another iterated integral can be defined by writing

$$G(x) = \int_c^d f(x, y) dy, \quad a \leq x \leq b,$$

Defining

$$I_2 = \int_a^b G(x) dx = \int_a^b \left(\int_c^d f(x, y) dy \right) dx,$$

which we usually write as

$$I_2 = \int_a^b dx \int_c^d f(x, y) dy.$$





$$S(\mathbf{P}) = \sum_{j=1}^k M_j V(R_j).$$

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3.9. Upper and Lower Integrals

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The *upper integral* of f over R , denoted by

$$\overline{\int}_R f(\mathbf{X}) d\mathbf{X},$$

is the infimum of all upper sums.

- **Upper and Lower Integrals:** The *lower sum* of f over $\mathbf{P}$ is

$$s(\mathbf{P}) = \sum_{j=1}^k m_j V(R_j).$$

The *lower integral* of f over R , denoted by

$$\underline{\int}_R f(\mathbf{X}) d\mathbf{X},$$

is the supremum of all lower sums.

Theorem: Let f be bounded on a rectangle R and let $\mathbf{P}$ be a partition of R .

Then

1. The upper sum $S(\mathbf{P})$ of f over $\mathbf{P}$ is the supremum of the set of all Riemann sums of f over $\mathbf{P}$.
2. The lower sum $s(\mathbf{P})$ of f over $\mathbf{P}$ is the infimum of the set of all Riemann sums of f over $\mathbf{P}$.

Remarks: If

$$m \leq f(\mathbf{X}) \leq M \quad \text{for } \mathbf{X} \text{ in } R,$$

then

$$mV(R) \leq s(\mathbf{P}) \leq S(\mathbf{P}) \leq MV(R);$$

therefore, $\overline{\int}_R f(\mathbf{X}) d\mathbf{X}$ and $\underline{\int}_R f(\mathbf{X}) d\mathbf{X}$ exist, are unique, and satisfy the inequalities

$$mV(R) \leq \overline{\int}_R f(\mathbf{X}) d\mathbf{X} \leq MV(R)$$

and

$$mV(R) \leq \underline{\int}_R f(\mathbf{X}) d\mathbf{X} \leq MV(R).$$

Remarks: The upper and lower integrals are also written as

$$\overline{\int}_R f(\mathbf{X}) d\mathbf{X} = \sup \{ S(\mathbf{P}) : \mathbf{P} \text{ is a partition of } R \}$$

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(see (3.18)), so (3.17) implies that

$$\left| \sigma - \sum_{i=1}^r \sum_{j=1}^s (\bar{x}_i + \bar{y}_j)(x_i - x_{i-1})(y_j - y_{j-1}) \right| \leq \|\mathbf{P}\|(b-a)(d-c). \quad (3.20)$$

It now follows that

$$\begin{aligned} & \sum_{i=1}^r \sum_{j=1}^s \bar{x}_i(x_i - x_{i-1})(y_j - y_{j-1}) \\ &= \left[\sum_{i=1}^r \bar{x}_i(x_i - x_{i-1}) \right] \left[\sum_{j=1}^s (y_j - y_{j-1}) \right] \\ &= (d-c) \sum_{i=1}^r \bar{x}_i(x_i - x_{i-1}) \quad (\text{from (3.18)}) \\ &= \frac{d-c}{2} \sum_{i=1}^r (x_i^2 - x_{i-1}^2) \quad (\text{from (3.14)}) \\ &= \frac{d-c}{2} (b^2 - a^2) \quad (\text{from (3.19)}). \end{aligned}$$

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Similarly,

$$\sum_{i=1}^r \sum_{j=1}^s \bar{y}_j(x_i - x_{i-1})(y_j - y_{j-1}) = \frac{b-a}{2}(d^2 - c^2).$$

Therefore, (3.20) can be written as

$$\left| \sigma - \frac{d-c}{2}(b^2 - a^2) - \frac{b-a}{2}(d^2 - c^2) \right| \leq \|\mathbf{P}\|(b-a)(d-c).$$

Since the right side can be made as small as we wish by choosing $\|\mathbf{P}\|$ sufficiently small,

$$\int_a^b (x+n) d(x-n) = \frac{1}{2} [(b-a)(b^2 - a^2) + (b-a)(d^2 - c^2)].$$

Theorem: If f is unbounded on the nondegenerate rectangle R in $\mathbb{R}^n$, then f is not integrable on R .

Proof: We will show that if f is unbounded on R , $\mathbf{P} = \{R_1, R_2, \dots, R_k\}$ is any partition of R , and $M > 0$, then there are Riemann sums σ and σ' for f over $\mathbf{P}$ such that

$$|\sigma - \sigma'| \geq M. \quad (3.21)$$

This implies that f cannot satisfy definition of Riemann integral (why?).

Let

$$\sigma = \sum_{j=1}^k (\mathbf{X}_j) V(R_j)$$

be a Riemann sum of f over $\mathbf{P}$. Let

$$\sigma' = \sum_{j=1}^k f(\mathbf{X}_j) V(R_j)$$

be a Riemann sum of f over $\mathbf{P}$.

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There must be an integer i in $\{1, 2, \dots, k\}$ such that





3.2 Absolute integrability

We say that f is *absolutely integrable* on $[a, b]$ if f is locally integrable on $[a, b]$ and $\int_a^b |f(x)| dx < \infty$. In this case we also say that $\int_a^b f(x) dx$ *converges absolutely* or is *absolutely convergent*.

Remark: If f is absolutely integrable on $[a, b]$, then f is absolutely integrable on $[a, b]$, since $|f| = f$.

Example: Since

$$\left| \frac{\sin x}{x^p} \right| \leq \frac{1}{x^p}$$

and $\int_1^\infty x^{-p} dx < \infty$ if $p > 1$.

The comparison theorem implies that

$$\int_1^\infty \frac{|\sin x|}{x^p} dx < \infty, \quad p > 1.$$

The function

$$f(x) = \frac{\sin x}{x^p}$$

is absolutely integrable on $[1, \infty)$ if $p > 1$.

Example: It is not absolutely integrable on $[1, \infty)$ if $p \leq 1$.

To see this, we first consider the case where $p = 1$.

Let k be an integer greater than 3. Then

$$\begin{aligned} \int_1^{k\pi} \frac{|\sin x|}{x} dx &> \int_\pi^{k\pi} \frac{|\sin x|}{x} dx \\ &= \sum_{j=1}^{k-1} \int_{j\pi}^{(j+1)\pi} \frac{|\sin x|}{x} dx \\ &> \sum_{j=1}^{k-1} \frac{1}{(j+1)\pi} \int_{j\pi}^{(j+1)\pi} |\sin x| dx. \end{aligned} \quad (3.5)$$

But

$$\int_{j\pi}^{(j+1)\pi} |\sin x| dx = \int_0^\pi \sin x dx = 2,$$

so (3.5) implies that

$$\int_1^{k\pi} \frac{|\sin x|}{x} dx > \frac{2}{\pi} \sum_{j=1}^{k-1} \frac{1}{j+1}. \quad (3.6)$$

However,

$$\frac{1}{j+1} \geq \int_{j+1}^{j+2} \frac{dx}{x}, \quad j = 1, 2, \dots,$$

so (3.6) implies that

$$\begin{aligned} \int_1^{k\pi} \frac{|\sin x|}{x} dx &> \frac{2}{\pi} \sum_{j=1}^{k-1} \int_{j+1}^{j+2} \frac{dx}{x} \\ &= \frac{2}{\pi} \int_2^{k+1} \frac{dx}{x} = \frac{2}{\pi} \log \frac{k+1}{2}. \end{aligned}$$

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From this, you can see that if $f(x) \geq 0$ on some subinterval $[a_1, b)$ of $[a, b)$, but not necessarily for all x in $[a, b)$, we can still use the convention introduced earlier for positive functions; that is, we can write $\int_a^b f(x) dx < \infty$ if the improper integral converges or $\int_a^b f(x) dx = \infty$ if it diverges.

Theorem: Suppose that f and g are locally integrable on $[a, b)$, $g(x) > 0$ and $f(x) \geq 0$ on some subinterval $[a_1, b)$ of $[a, b)$, and

$$\lim_{x \rightarrow b^-} \frac{f(x)}{g(x)} = M. \quad (3.2)$$

- If $0 < M < \infty$, then $\int_a^b f(x) dx$ and $\int_a^b g(x) dx$ converge or diverge together.
- If $M = \infty$ and $\int_a^b g(x) dx = \infty$, then $\int_a^b f(x) dx = \infty$.
- If $M = 0$ and $\int_a^b g(x) dx < \infty$, then $\int_a^b f(x) dx < \infty$.

Proof: From (3.3), there is a point a_2 in $[a_1, b)$ such that

$$0 < \frac{M}{2} < \frac{f(x)}{g(x)} < \frac{3M}{2}, \quad a_2 \leq x < b,$$

and therefore

$$\frac{M}{2} g(x) < f(x) < \frac{3M}{2} g(x), \quad a_2 \leq x < b. \quad (3.4)$$

The first inequality in (3.4) imply that

$$\int_{a_2}^b g(x) dx < \infty \quad \text{if} \quad \int_{a_2}^b f(x) dx < \infty.$$

The second inequality in (3.4) imply that

$$\int_{a_2}^b f(x) dx < \infty \quad \text{if} \quad \int_{a_2}^b g(x) dx < \infty.$$

Therefore, $\int_{a_2}^b f(x) dx$ and $\int_{a_2}^b g(x) dx$ converge or diverge together, and in the latter case they must diverge to ∞ , since their integrands are nonnegative. If $M = \infty$, there is a point a_2 in $[a_1, b)$ such that

$$f(x) \geq g(x), \quad a_2 \leq x \leq b,$$

We have $\int_a^b f(x) dx = \infty$. If $M = 0$, there is a point a_2 in $[a_1, b)$ such that

$$f(x) \leq g(x), \quad a_2 \leq x \leq b,$$

so we have $\int_a^b f(x) dx < \infty$.





2.23.1 Positive Definite

Let r be a positive integer and $\mathbf{X}_0 = (x_{10}, x_{20}, \dots, x_{n0})$. A function of the form

$$p(\mathbf{X}) = \sum_r a_{r_1 r_2 \dots r_n} (x_1 - x_{10})^{r_1} (x_2 - x_{20})^{r_2} \dots (x_n - x_{n0})^{r_n}, \quad (2.102)$$

where the coefficients $\{a_{r_1 r_2 \dots r_n}\}$ are constants and the summation is over all n -tuples of nonnegative integers $(r_1, r_2, \dots, r_n)$ such that

$$r_1 + r_2 + \dots + r_n = r,$$

is a homogeneous polynomial of degree r in $\mathbf{X} - \mathbf{X}_0$, provided that at least one of the coefficients is nonzero. For example, if f satisfies the conditions of r th differential, then the function

$$p(\mathbf{X}) = (d_{\mathbf{X}_0}^{(r)} f)(\mathbf{X} - \mathbf{X}_0)$$

is such a polynomial if at least one of the r th-order mixed partial derivatives of f at $\mathbf{X}_0$ is nonzero. Clearly, $p(\mathbf{X}_0) = 0$ if p is a homogeneous polynomial of degree $r \geq 1$ in $\mathbf{X} - \mathbf{X}_0$.

If $p(\mathbf{X}) \geq 0$ for all $\mathbf{X}$, we say that p is *positive semidefinite*; if $p(\mathbf{X}) > 0$ except when $\mathbf{X} = \mathbf{X}_0$, p is *positive definite*. Similarly, p is *negative semidefinite* if $p(\mathbf{X}) \leq 0$ or *negative definite* if $p(\mathbf{X}) < 0$ for all $\mathbf{X} \neq \mathbf{X}_0$. In all these cases, p is *semidefinite*. With p as in (2.102),

$$p(-\mathbf{X} + 2\mathbf{X}_0) = (-1)^r p(\mathbf{X}),$$

so p cannot be semidefinite if r is odd.

Example: The polynomial

$$p(x, y, z) = x^2 + y^2 + z^2 + xy + xz + yz$$

is homogeneous of degree 2 in $\mathbf{X} = (x, y, z)$. We can rewrite p as

$$p(x, y, z) = \frac{1}{2} [(x+y)^2 + (y+z)^2 + (z+x)^2].$$

so p is nonnegative, and $p(\bar{x}, \bar{y}, \bar{z}) = 0$ if and only if

$$x + y = \bar{y} + \bar{z} = \bar{z} + \bar{x} = 0,$$

which is equivalent to $(\bar{x}, \bar{y}, \bar{z}) = (0, 0, 0)$. Therefore, p is positive definite and $-p$ is negative definite.

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2.23. Taylor's Theorem for Functions of n Variables

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Example: The polynomial

$$p_1(x, y, z) = x^2 + y^2 + z^2 + 2xy$$

$$p_1(x, y, z) = (x+y)^2 + z^2,$$

so p_1 is nonnegative. Since $p_1(1, -1, 0) = 0$, p_1 is positive semidefinite and $-p_1$ is negative semidefinite.

The polynomial





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Substituting this into (2.82) yields

$$\frac{\partial^2 h}{\partial u_k \partial u_l} = \sum_{j=1}^n \frac{\partial f}{\partial x_j} \frac{\partial^2 x_j}{\partial u_k \partial u_l} + \sum_{j=1}^n \frac{\partial x_j}{\partial u_l} \sum_{s=1}^n \frac{\partial^2 f}{\partial x_s \partial x_j} \frac{\partial x_s}{\partial u_k}. \quad (2.84)$$

To compute $h_{u_l u_k}(\mathbf{U}_0)$ from this formula, we evaluate the partial derivatives of $x_1, x_2, \dots, x_n$ at $\mathbf{U}_0$ and those of f at $\mathbf{X}_0 = \mathbf{X}(\mathbf{U}_0)$. The formula is valid if $x_1, x_2, \dots, x_n$ and their first partial derivatives are differentiable at $\mathbf{U}_0$ and $f, f_{x_1}, f_{x_2}, \dots, f_{x_n}$ and their first partial derivatives are differentiable at $\mathbf{X}_0$.

Example: Let (r, θ) be polar coordinates in the xy -plane; that is,

$$x = r \cos \theta, \quad y = r \sin \theta.$$

Suppose that $f = f(x, y)$ is differentiable on a set S , and let

$$h(r, \theta) = f(r \cos \theta, r \sin \theta).$$

We have

$$\begin{aligned} \frac{\partial h}{\partial r} &= \frac{\partial f}{\partial x} \frac{\partial x}{\partial r} + \frac{\partial f}{\partial y} \frac{\partial y}{\partial r} = \cos \theta \frac{\partial f}{\partial x} + \sin \theta \frac{\partial f}{\partial y} \\ \frac{\partial h}{\partial \theta} &= \frac{\partial f}{\partial x} \frac{\partial x}{\partial \theta} + \frac{\partial f}{\partial y} \frac{\partial y}{\partial \theta} = -r \sin \theta \frac{\partial f}{\partial x} + r \cos \theta \frac{\partial f}{\partial y}, \end{aligned} \quad (2.85)$$

where f_x and f_y are evaluated at $(x, y) = (r \cos \theta, r \sin \theta)$.

Example: Suppose that f_x and f_y just calculated are differentiable on an open set S in $\mathbb{R}^2$. Differentiating (2.85) with respect to r yields

$$\begin{aligned} \frac{\partial^2 h}{\partial r^2} &= \cos \theta \frac{\partial}{\partial r} \left(\frac{\partial f}{\partial x} \right) + \sin \theta \frac{\partial}{\partial r} \left(\frac{\partial f}{\partial y} \right) \\ &= \cos \theta \left(\frac{\partial^2 f}{\partial x^2} \frac{\partial x}{\partial r} + \frac{\partial^2 f}{\partial y \partial x} \frac{\partial y}{\partial r} \right) + \sin \theta \left(\frac{\partial^2 f}{\partial x \partial y} \frac{\partial x}{\partial r} + \frac{\partial^2 f}{\partial y^2} \frac{\partial y}{\partial r} \right). \end{aligned} \quad (2.86)$$

if $(x, y) \in S$. Since

$$\frac{\partial x}{\partial r} = \cos \theta, \quad \frac{\partial y}{\partial r} = \sin \theta, \quad \text{and} \quad \frac{\partial^2 f}{\partial x \partial y} = \frac{\partial^2 f}{\partial y \partial x}$$

if $(x, y) \in S$. The equation (2.86) yields

$$\frac{\partial^2 h}{\partial r^2} = \cos^2 \theta \frac{\partial^2 f}{\partial x^2} + 2 \sin \theta \cos \theta \frac{\partial^2 f}{\partial x \partial y} + \sin^2 \theta \frac{\partial^2 f}{\partial y^2}.$$

Differentiating (2.85) with respect to θ yields

$$\begin{aligned} \frac{\partial^2 h}{\partial \theta \partial r} &= -\sin \theta \frac{\partial f}{\partial x} + \cos \theta \frac{\partial f}{\partial y} + \cos \theta \frac{\partial}{\partial \theta} \left(\frac{\partial f}{\partial x} \right) + \sin \theta \frac{\partial}{\partial \theta} \left(\frac{\partial f}{\partial y} \right) \\ &= -\sin \theta \frac{\partial f}{\partial x} + \cos \theta \frac{\partial f}{\partial y} + \cos \theta \left(\frac{\partial^2 f}{\partial x^2} \frac{\partial x}{\partial \theta} + \frac{\partial^2 f}{\partial y \partial x} \frac{\partial y}{\partial \theta} \right) \\ &\quad + \sin \theta \left(\frac{\partial^2 f}{\partial x \partial y} \frac{\partial x}{\partial \theta} + \frac{\partial^2 f}{\partial y^2} \frac{\partial y}{\partial \theta} \right). \end{aligned}$$

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$$f(\mathbf{X}_j) - f(\mathbf{X}_{j-1}) = f_{x_j}(\mathbf{X}_j)(x_j - x_{j0}),$$

and (2.53) implies that

$$\begin{aligned} f(\mathbf{X}) - f(\mathbf{X}_0) &= \sum_{j=1}^n f_{x_j}(\hat{\mathbf{X}}_j)(x_j - x_{j0}) \\ &= \sum_{j=1}^n f_{x_j}(\mathbf{X}_0)(x_j - x_{j0}) + \sum_{j=1}^n [f_{x_j}(\hat{\mathbf{X}}_j) - f_{x_j}(\mathbf{X}_0)](x_j - x_{j0}). \end{aligned}$$

From this and (2.52),

$$\left| f(\mathbf{X}) - f(\mathbf{X}_0) - \sum_{j=1}^n f_{x_j}(\mathbf{X}_0)(x_j - x_{j0}) \right| \leq \varepsilon \sum_{j=1}^n |x_j - x_{j0}| \leq n\varepsilon |\mathbf{X} - \mathbf{X}_0|,$$

which implies that f is differentiable at $\mathbf{X}_0$.

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2.17.4 Continuously Differentiable Function

We say that f is *continuously differentiable* on a subset S of $\mathbb{R}^n$ if S is contained in an open set on which $f_{x_1}, f_{x_2}, \dots, f_{x_n}$ are continuous.

The above theorem implies that such a function is differentiable at each $\mathbf{X}_0$ in S .

Example: If

$$f(x, y) = \frac{x^2 + y^2}{x - y},$$

then

$$f_x(x, y) = \frac{2x}{x - y} - \frac{x^2 + y^2}{(x - y)^2}$$

$$f_y(x, y) = \frac{2y}{x - y} + \frac{x^2 + y^2}{(x - y)^2}.$$

Since f_x and f_y are continuous on

$$S = \{(x, y) : x \neq y\}$$

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2.17. Differentiability of Functions of Several Variables

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f is continuously differentiable on S .

Remark: If $f_{x_1}, f_{x_2}, \dots, f_{x_n}$ exist on a neighborhood of $\mathbf{X}_0$ and are continuous at $\mathbf{X}_0$, then f is differentiable at $\mathbf{X}_0$. These conditions are not necessary for differentiability; that is, a function may be differentiable at a point $\mathbf{X}_0$ even if its first partial derivatives are not continuous at $\mathbf{X}_0$.

Example: let

$$f(x, y) = \begin{cases} (x - y)^2 \sin \frac{1}{x - y}, & x \neq y, \\ 0, & x = y. \end{cases}$$

Then





$$f(\mathbf{X}_j) - f(\mathbf{X}_{j-1}) = f_{x_j}(\mathbf{X}_j)(x_j - x_{j0}),$$

and (2.53) implies that

$$\begin{aligned} f(\mathbf{X}) - f(\mathbf{X}_0) &= \sum_{j=1}^n f_{x_j}(\hat{\mathbf{X}}_j)(x_j - x_{j0}) \\ &= \sum_{j=1}^n f_{x_j}(\mathbf{X}_0)(x_j - x_{j0}) + \sum_{j=1}^n [f_{x_j}(\hat{\mathbf{X}}_j) - f_{x_j}(\mathbf{X}_0)](x_j - x_{j0}). \end{aligned}$$

From this and (2.52),

$$\left| f(\mathbf{X}) - f(\mathbf{X}_0) - \sum_{j=1}^n f_{x_j}(\mathbf{X}_0)(x_j - x_{j0}) \right| \leq \varepsilon \sum_{j=1}^n |x_j - x_{j0}| \leq n\varepsilon |\mathbf{X} - \mathbf{X}_0|,$$

which implies that f is differentiable at $\mathbf{X}_0$.

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The above theorem implies that such a function is differentiable at each $\mathbf{X}_0$ in S .

Example: If

$$f(x, y) = \frac{x^2 + y^2}{x - y},$$

then

$$f_x(x, y) = \frac{2x}{x - y} - \frac{x^2 + y^2}{(x - y)^2}$$

$$f_y(x, y) = \frac{2y}{x - y} + \frac{x^2 + y^2}{(x - y)^2}.$$

Since f_x and f_y are continuous on

$$S = \{(x, y) : x \neq y\}$$

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f is continuously differentiable on S .

Remark: If $f_{x_1}, f_{x_2}, \dots, f_{x_n}$ exist on a neighborhood of $\mathbf{X}_0$ and are continuous at $\mathbf{X}_0$, then f is differentiable at $\mathbf{X}_0$. These conditions are not necessary for differentiability; that is, a function may be differentiable at a point $\mathbf{X}_0$ even if its first partial derivatives are not continuous at $\mathbf{X}_0$.

Example: let

$$f(x, y) = \begin{cases} (x - y)^2 \sin \frac{1}{x - y}, & x \neq y, \\ 0, & x = y. \end{cases}$$

Then



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$$\lim_{\mathbf{X} \rightarrow \mathbf{X}_0} \frac{f(\mathbf{X}) - f(\mathbf{X}_0) - \sum_{i=1}^n f_{x_i}(\mathbf{X}_0)(x_i - x_{i0})}{|\mathbf{X} - \mathbf{X}_0|} = 0.$$

Proof: Let i be a given integer in $\{1, 2, \dots, n\}$. Let $\mathbf{X} = \mathbf{X}_0 + t\mathbf{E}_i$, so that $x_i = x_{i0} + t$, $x_j = x_{j0}$ if $j \neq i$, and $|\mathbf{X} - \mathbf{X}_0| = |t|$.

Then

$$\lim_{\mathbf{X} \rightarrow \mathbf{X}_0} \frac{f(\mathbf{X}) - f(\mathbf{X}_0) - \sum_{i=1}^n m_i(x_i - x_{i0})}{|\mathbf{X} - \mathbf{X}_0|} = 0.$$

and the differentiability of f at $\mathbf{X}_0$ imply that

$$\lim_{t \rightarrow 0} \frac{f(\mathbf{X}_0 + t\mathbf{E}_i) - f(\mathbf{X}_0) - m_i t}{t} = 0.$$

Hence,

$$\lim_{t \rightarrow 0} \frac{f(\mathbf{X}_0 + t\mathbf{E}_i) - f(\mathbf{X}_0) - m_i t}{t} = 0.$$

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This proves (2.46), since the limit on the left is $f_{x_i}(\mathbf{X}_0)$, by definition.

2.17.1 Linear Function

A *linear function* is a function of the form

$$L(\mathbf{X}) = m_1 x_1 + m_2 x_2 + \dots + m_n x_n, \quad (2.47)$$

where $m_1, m_2, \dots, m_n$ are constants. From definition of differentiability, f is differentiable at $\mathbf{X}_0$ if and only if there is a linear function L such that $f(\mathbf{X}) - f(\mathbf{X}_0)$ can be approximated so well near $\mathbf{X}_0$ by

$$L(\mathbf{X}) - L(\mathbf{X}_0) = L(\mathbf{X} - \mathbf{X}_0)$$

that

$$f(\mathbf{X}) - f(\mathbf{X}_0) = L(\mathbf{X} - \mathbf{X}_0) + E(\mathbf{X})(|\mathbf{X} - \mathbf{X}_0|), \quad (2.48)$$

where

$$\lim_{\mathbf{X} \rightarrow \mathbf{X}_0} E(\mathbf{X}) = 0. \quad (2.49)$$

Theorem: If f is differentiable at $\mathbf{X}_0$, then f is continuous at $\mathbf{X}_0$.

Proof: From $L(\mathbf{X}) = m_1 x_1 + m_2 x_2 + \dots + m_n x_n$, and Schwarz's inequality,

$$|L(\mathbf{X} - \mathbf{X}_0)| \leq M|\mathbf{X} - \mathbf{X}_0|$$

where

$$M = (m_1^2 + m_2^2 + \dots + m_n^2)^{1/2}.$$

This and $f(\mathbf{X}) - f(\mathbf{X}_0) = L(\mathbf{X} - \mathbf{X}_0) + E(\mathbf{X})(|\mathbf{X} - \mathbf{X}_0|)$, imply that

$$|f(\mathbf{X}) - f(\mathbf{X}_0)| \leq (M + |E(\mathbf{X})|)|\mathbf{X} - \mathbf{X}_0|,$$

which, with (2.49), implies that f is continuous at $\mathbf{X}_0$.

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2.17.2 Differential

The linear function



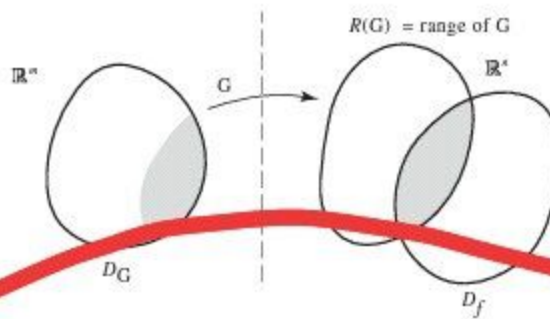


Figure 2.7: Composite of vector valued functions

Theorem: Suppose that U_0 is in T and is a limit point of T , G is continuous at U_0 , and f is continuous at $X_0 = G(U_0)$. Then $h = f \circ G$ is continuous at U_0 .

Proof: Suppose that $\varepsilon > 0$. Since f is continuous at $X_0 = G(U_0)$, there is an $\varepsilon_1 > 0$ such that

$$|f(X) - f(G(U_0))| < \varepsilon \quad (2.30)$$

if

$$|X - G(U_0)| < \varepsilon_1 \quad \text{and} \quad X \in D_f. \quad (2.31)$$

Since G is continuous at U_0 , there is a $\delta > 0$ such that

$$|G(U) - G(U_0)| < \varepsilon_1 \quad \text{if} \quad |U - U_0| < \delta \quad \text{and} \quad U \in D_G.$$

By taking $X = G(U)$ in (2.30) and (2.31), we see that

$$|h(U) - h(U_0)| = |f(G(U)) - f(G(U_0))| < \varepsilon$$

if

$$|U - U_0| < \delta \quad \text{and} \quad U \in T.$$

Example: If

$$f(s) = \sqrt{s}$$

and

$$g(x, y) = 1 - x^2 - 2y^2,$$

then $D_f = [0, \infty]$, $D_g = \mathbb{R}^2$, and

$$T = \{(x, y) : x^2 + 2y^2 \leq 1\}.$$

We have proved that g is continuous on $\mathbb{R}^2$.

We can obtain the same conclusion by observing that the functions $p_1(x, y) = x$ and $p_2(x, y) = y$ are continuous on $\mathbb{R}^2$.

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to make it continuous at $(0, 0)$.

The limit

$$\lim_{(x,y) \rightarrow (0,0)} f(x,y)$$

does not exist.

Consequently, it is impossible to define the function at origin to make it continuous.

Theorem: If f and g are continuous on a set S in $\mathbb{R}^n$, then so are $f + g$, $f - g$, and fg . Also, f/g is continuous at each $\mathbf{X}_0$ in S such that $g(\mathbf{X}_0) \neq 0$.

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2.14. Vector Valued Functions

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2.14.1 Vector Valued Functions

Suppose that $g_1, g_2, \dots, g_n$ are real-valued functions defined on a subset T of $\mathbb{R}^m$.

We define the *vector-valued function* $\mathbf{G}$ on T by

$$\mathbf{G}(\mathbf{U}) = (g_1(\mathbf{U}), g_2(\mathbf{U}), \dots, g_n(\mathbf{U})), \quad \mathbf{U} \in T.$$

Then $g_1, g_2, \dots, g_n$ are the *component functions* of $\mathbf{G} = (g_1, g_2, \dots, g_n)$. We say that

$$\lim_{\mathbf{U} \rightarrow \mathbf{U}_0} \mathbf{G}(\mathbf{U}) = \mathbf{L} = (L_1, L_2, \dots, L_n)$$

if

$$\lim_{\mathbf{U} \rightarrow \mathbf{U}_0} g_i(\mathbf{U}) = L_i, \quad 1 \leq i \leq n,$$

and that $\mathbf{G}$ is *continuous* at $\mathbf{U}_0$ if $g_1, g_2, \dots, g_n$ are each continuous at $\mathbf{U}_0$.

Theorem: For a vector-valued function $\mathbf{G}$,

$$\lim_{\mathbf{U} \rightarrow \mathbf{U}_0} \mathbf{G}(\mathbf{U}) = \mathbf{L}$$

if and only if for each $\varepsilon > 0$ there is a $\delta > 0$ such that

$$|\mathbf{G}(\mathbf{U}) - \mathbf{L}| < \varepsilon \quad \text{whenever} \quad 0 < |\mathbf{U} - \mathbf{U}_0| < \delta \quad \text{and} \quad \mathbf{U} \in D_{\mathbf{G}}.$$

Similarly, $\mathbf{G}$ is continuous at $\mathbf{U}_0$ if and only if for each $\varepsilon > 0$ there is a $\delta > 0$ such that

$$|\mathbf{G}(\mathbf{U}) - \mathbf{G}(\mathbf{U}_0)| < \varepsilon \quad \text{whenever} \quad |\mathbf{U} - \mathbf{U}_0| < \delta \quad \text{and} \quad \mathbf{U} \in D_{\mathbf{G}}.$$

2.14.1 Composite Function

Let f be a real-valued function defined on a subset of $\mathbb{R}^n$, and let the vector-valued function $\mathbf{G} = (g_1, g_2, \dots, g_n)$ be defined on a domain $D_{\mathbf{G}}$ in $\mathbb{R}^m$.

Let the set

$$T = \{\mathbf{U} : \mathbf{U} \in D_{\mathbf{G}} \quad \text{and} \quad \mathbf{G}(\mathbf{U}) \in D_f\},$$

be nonempty.

Composite function: Define the real-valued composite function





Therefore, the assumption that there is a polygonal path in S from $\mathbf{X}_1$ to $\mathbf{X}_2$ must be false.

For necessity, suppose that S is a connected open set and $\mathbf{X}_0 \in S$. Let A be the set consisting of $\mathbf{X}_0$ and the points in S that can be connected to $\mathbf{X}_0$ by polygonal paths in S . Let B be the set of points in S that cannot be connected to $\mathbf{X}_0$ by polygonal paths. If $\mathbf{Y}_0 \in S$, then S contains an ε -neighborhood $N_\varepsilon(\mathbf{Y}_0)$ of $\mathbf{Y}_0$, since S is open. Any point $\mathbf{Y}_1$ in $N_\varepsilon(\mathbf{Y}_0)$ can be connected to $\mathbf{Y}_0$ by the line segment

$$\mathbf{X} = t\mathbf{Y}_1 + (1 - t)\mathbf{Y}_0, \quad 0 \leq t \leq 1,$$

which lies in $N_\varepsilon(\mathbf{Y}_0)$ and therefore in S . This implies that $\mathbf{Y}_0$ can be connected to $\mathbf{X}_0$ by a polygonal path in S if and only if every member of $N_\varepsilon(\mathbf{Y}_0)$ can also. Thus, $N_\varepsilon(\mathbf{Y}_0) \subset A$ if $\mathbf{Y}_0 \in A$, and $N_\varepsilon(\mathbf{Y}_0) \subset B$ if $\mathbf{Y}_0 \in B$. Therefore, A and B are open. Since $A \cap B = \emptyset$, this implies that $A \cap \overline{B} = \overline{A} \cap B = \emptyset$. Since A is nonempty ($\mathbf{X}_0 \in A$), it now follows that $B = \emptyset$, since if $B \neq \emptyset$, S would be disconnected. Therefore, $A = S$, which completes the proof of necessity.

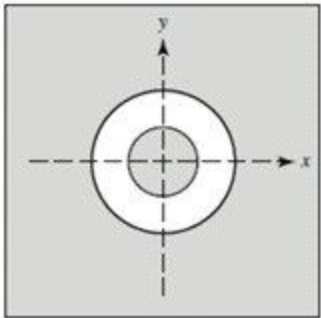
Remark: Any polygonally connected set, open or not, is connected. The converse is false. A set (not open) may be disconnected.

Regions in $\mathbb{R}^n$: A region S in $\mathbb{R}^n$ is a set with some, all, or none of its boundary; the boundary of S is a limit point of S^0 .

Example: Intervals are the only regions in $\mathbb{R}$. The n -ball $B_r(\mathbf{X}_0)$ is a region in $\mathbb{R}^n$, as is its closure $\overline{B}_r(\mathbf{X}_0)$. The set $S = \{(x, y) : x^2 + y^2 \leq 1 \text{ or } x^2 + y^2 \geq 4\}$ is not a region in $\mathbb{R}^2$, since it is not connected.

The set S_1 obtained by adding the line segment

$$L_1: \mathbf{X} = t(0, 2) + (1 - t)(0, 1), \quad 0 < t < 1,$$





disjoint nonempty sets such that neither contains a limit point of the other.

If S cannot be expressed as $S = A \cup B$, where

$$A \neq \emptyset, \quad B \neq \emptyset, \quad \overline{A} \cap B = \emptyset, \quad \text{and} \quad A \cap \overline{B} = \emptyset. \quad (2.13)$$

If S can be expressed in this way, then S is *disconnected*.

Example: The empty set and singleton sets are connected, because they cannot be represented as the union of two disjoint nonempty sets.

Example: The space $\mathbb{R}^n$ is connected.

If $\mathbb{R}^n = A \cup B$ with $\overline{A} \cap B = \emptyset$ and $A \cap \overline{B} = \emptyset$, then $\overline{A} \subset A$ and $\overline{B} \subset B$.

That is, A and B are both closed and therefore are both open.

Since the only nonempty subset of $\mathbb{R}^n$ that is both open and closed is $\mathbb{R}^n$ itself, one of A and B is $\mathbb{R}^n$ and the other is empty.

2.7.1 Polygonal Path

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If $\mathbf{X}_1, \mathbf{X}_2, \dots, \mathbf{X}_k$ are points in $\mathbb{R}^n$

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Let L_i is the line segment from $\mathbf{X}_i$ to $\mathbf{X}_{i+1}$, $1 \leq i \leq k-1$, we say that $L_1, L_2, \dots, L_{k-1}$ form a *polygonal path* from $\mathbf{X}_1$ to $\mathbf{X}_k$.

We say that $\mathbf{X}_1$ and $\mathbf{X}_k$ are *connected* by the polygonal path.

2.8 Polygonally Connected Set

A set S is *polygonally connected* if every pair of points in S can be connected by a polygonal path lying entirely in S .

Theorem: An open set S in $\mathbb{R}^n$ is connected if and only if it is polygonally connected.

Proof: For sufficiency, we will show that if S is disconnected, then S is not polygonally connected.

Let $S = A \cup B$, where A and B satisfy

$$A \neq \emptyset, \quad B \neq \emptyset, \quad \overline{A} \cap B = \emptyset, \quad \text{and} \quad A \cap \overline{B} = \emptyset.$$

2.8. Polygonally Connected Set

Suppose that $\mathbf{X}_1 \in A$ and $\mathbf{X}_2 \in B$, and assume that there is a polygonal path in S connecting $\mathbf{X}_1$ to $\mathbf{X}_2$. Then some line segment L in this path must contain a point $\mathbf{Y}_1$ in A and a point $\mathbf{Y}_2$ in B .

The line segment



3.12. Differentiable Surfaces

Q1- S_1, S_2 has zero content then prove that $S_1 \cup S_2$ has zero content.

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mcq/s

Theorem: If f and g are integrable on S and $f(\mathbf{X}) \leq g(\mathbf{X})$ for $\mathbf{X}$ in S , then

$$\int_S f(\mathbf{X}) d\mathbf{X} \leq \int_S g(\mathbf{X}) d\mathbf{X}.$$

$$\left| \int_S f(\mathbf{X}) d\mathbf{X} \right| \leq \int_S |f(\mathbf{X})| d\mathbf{X}.$$

Theorem: If f and g are integrable on S , then so is the product fg .

(3.22) implies (3.21).

$$|\sigma - \sigma'| = |f(\mathbf{X}) - f(\mathbf{X}_i)|V(R_i),$$

3.9 Upper and Lower Integrals

If f is bounded on a rectangle R in $\mathbb{R}^n$ and $\mathbf{P} = \{R_1, R_2, \dots, R_k\}$ is a partition of R .

Let

$$M_j = \sup_{\mathbf{X} \in R_j} f(\mathbf{X}), \quad m_j = \inf_{\mathbf{X} \in R_j} f(\mathbf{X}).$$

The upper sum of f over $\mathbf{P}$ is

$$S(\mathbf{P}) = \sum_{j=1}^k M_j V(R_j).$$

Definition ?? yields

$$\int_0^1 \log x \, dx = -1.$$

The function $f(x) = \cos x$ is locally integrable on $[0, \infty)$ and

$$\lim_{c \rightarrow \infty} \int_0^c \cos x \, dx = \lim_{c \rightarrow \infty} \sin c$$

does not exist; thus, $\int_0^\infty \cos x \, dx$ diverges, but not to $\pm\infty$.

In connection with Definition ??, it is important to recognize that the improper integrals $\int_a^\infty f(x) \, dx$ and $\int_a^b f(x) \, dx$ must converge separately for $\int_a^b f(x) \, dx$ to converge. For example, the existence of the symmetric limit

$$\lim_{R \rightarrow \infty} \int_{-R}^R f(x) \, dx,$$

which is called the *principal value* of $\int_{-\infty}^\infty f(x) \, dx$, does not imply that $\int_{-\infty}^\infty f(x) \, dx$ converges; thus,

$$\lim_{R \rightarrow \infty} \int_{-R}^R x \, dx = \lim_{R \rightarrow \infty} 0 = 0,$$

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If $\lim_{|(x,y,z)| \rightarrow \infty} f(x,y,z) = 1$ then prove that $f(x) = \cos\left(\frac{1}{x+2y+z}\right)$ is differential.

2.17. Differentiability of Functions of Several Variables

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2.17.2 Differential

The linear function

$$L(\mathbf{X}) = f_{x_1}(\mathbf{X}_0)x_1 + f_{x_2}(\mathbf{X}_0)x_2 + \cdots + f_{x_n}(\mathbf{X}_0)x_n.$$

This function is called the *differential of f at $\mathbf{X}_0$* . We will denote it by $d_{\mathbf{X}_0}f$ and its value by $(d_{\mathbf{X}_0}f)(\mathbf{X})$.

Thus,

$$(d_{\mathbf{X}_0}f)(\mathbf{X}) = f_{x_1}(\mathbf{X}_0)x_1 + f_{x_2}(\mathbf{X}_0)x_2 + \cdots + f_{x_n}(\mathbf{X}_0)x_n. \quad (2.50)$$

Real Analysis

MCQs = 20 Marks

- ① Prove that $d(S) = d(\bar{S})$ for any set in $\mathbb{R}^n$ — (5) marks
- ② Define locally integrable function and also show that $f(x) = \log x$ is locally integrable on $(0,1)$ — (5) marks
- ③ if $h(u,v) = f(u/v)$ then prove $u_h + u_{hv} = 0$ — (2) marks
- ④ Prove that S_1 and S_2 have zero content then $S_1 \cup S_2$ has zero content — (5) marks
- ⑤ Find upper and lower integral in $\mathbb{R}^n$ for a partition $P = \{R_1, R_2, \dots, R_k\}$ of R — (10) marks
- ⑥ let $h(u,v) = f(g_1(u,v), g_2(u,v))$ then find u_h and then write h explicitly as a function of u where

$$f(x,y) = 3x^2 + 4xy^2 + 3x,$$

$$g_1(u,v) = ve^{u+v-1},$$

$$g_2(u,v) = e^{-u+v-1}, \quad (u_0, v_0) = (0,1)$$

3.6. Rectangles in $\mathbb{R}^n$

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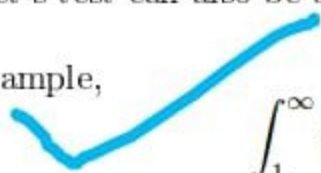
where the integral on the right converges absolutely.

Remark: Dirichlet's test is useful only if f is oscillatory at $b-$, since it can be shown that if f is nonoscillatory at $b-$ and F is bounded, then $\int_a^b f(x)g(x)dx < \infty$ if only g is locally integrable and bounded.

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Remark: Dirichlet's test can also be used to show that certain integrals diverge.

Example: For example,



$$\int_1^{\infty} x^q \sin x \, dx$$

most important Long Ques

diverges if $q > 0$, but none of the other tests that we have studied so far implies this. It is not enough to argue that the integrand does not approach zero as $x \rightarrow \infty$ (a common mistake), since this does not imply divergence. To see that the integral diverges, we observe that if it converged for some $q > 0$, then $F(x) = \int_1^x x^q \sin x \, dx$ would be bounded on $[1, \infty)$.

We could let

$$f(x) = x^q \sin x \quad \text{and} \quad g(x) = x^{-q}$$

in Dirichlet's test and conclude that

$$\int_1^{\infty} \sin x \, dx$$

also converges. This is false.

3.6 Rectangles in $\mathbb{R}^n$

The

$$S_1 \times S_2 \times \cdots \times S_n$$

of subsets $S_1, S_2, \dots, S_n$ of $\mathbb{R}$ is the set of points $(x_1, x_2, \dots, x_n)$ in $\mathbb{R}^n$ such that $x_1 \in S_1, x_2 \in S_2, \dots, x_n \in S_n$. For example, the Cartesian product of the two closed intervals

$$[a_1, b_1] \times [a_2, b_2] = \{(x, y) : a_1 \leq x \leq b_1, a_2 \leq y \leq b_2\}$$

is a rectangle in $\mathbb{R}^2$ with sides parallel to the x - and y -axes.

The Cartesian product of three closed intervals

$$\begin{aligned} [a_1, b_1] \times [a_2, b_2] \times [a_3, b_3] &= \{(x, y, z) : a_1 \leq x \leq b_1, \\ &\quad a_2 \leq y \leq b_2, a_3 \leq z \leq b_3\} \end{aligned}$$

is a rectangular parallelepiped in $\mathbb{R}^3$ with faces parallel to the coordinate axes. A *coordinate rectangle* R in $\mathbb{R}^n$ is the Cartesian product of n closed intervals; that is,

$$R = [a_1, b_1] \times [a_2, b_2] \times \cdots \times [a_n, b_n].$$

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Time Left 31 sec(s)

Quiz Start Time: 03:53 PM, 30 July 2018

Question # 1 of 10 (Start time: 03:53:17 PM, 30 July 2018) Total Marks: 1

and $f_{yz}(x_0, y_0)$ exists. Then,

Select the correct option

Reload Math Equations

- ☒ $f_{yz}(x_0, y_0) = f_{xy}(x_0, y_0)$
- ☐ $f_{xx}(x_0, y_0) = f_{yy}(x_0, y_0)$
- ☐ $f_x(x_0, y_0) = f_y(x_0, y_0)$
- ☐ None of these

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Question # 3 of 10 (Start time: 03:55:28 PM, 30 July 2018) Total Marks: 1

A critical point at which a function attains its maximum value among all points where it is defined is called

Select the correct option

- ☐ global minimum
- ☐ infimum
- ☐ global maximum
- ☐ None of these

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Question # 4 of 10 (Start time: 03:56:48 PM, 30 July 2018) Total Marks: 1

Which expression represents the differential of a function of n variables at the point X_0 ?

Select the correct option

Reload Math Equations

- ☐ $L(X) = f_{x_1}(X_0) + f_{x_2}(X_0) + \dots + f_{x_n}(X_0)$
- ☐ $L(X) = f(X_0)x_1 + f(X_0)x_2 + \dots + f(X_0)x_n$
- ☐ $L(X) = f_{x_1}(X_0)x_1 + f_{x_2}(X_0)x_2 + \dots + f_{x_n}(X_0)x_n$
- ☐ None of these

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Question # 5 of 10 (Start time: 03:57:57 PM, 30 July 2018) Total Marks: 1

Which statement(s) is(are) true about the function

$$f(x, y) = \frac{x^2 + y^2}{x - y}$$

Select the correct option Reload Math Equations

- | | |
|-----------------------|--|
| ☐ | $f_x(x, y) = \frac{2x}{x - y} - \frac{x^2 + y^2}{(x + y)^2}$ |
| ☐ | $f_y(x, y) = \frac{y}{x - y} + \frac{x^2 + y^2}{(x - y)^2}$ |
| ☐ | $f_y(x, y), f_x(x, y)$ are continuous everywhere. |
| ☐ | is differentiable every where except at the points where $y = x$ |

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Question # 7 of 10 (Start time: 03:59:57 PM, 30 July 2018) Total Marks: 1

Identify the false statement(s)

Select the correct option

Reload Math Equations

- ☐ The partial derivative is a special case of directional derivative.
- ☐ $\frac{\partial f(\mathbf{X})}{\partial \mathbf{E}_2}$
- ☒ with respect to third variable is $f_z = 2z$
- ☐ None of these

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Time Left 75 sec(s)

Quiz Start Time: 03:53 PM, 30 July 2018

Question # 8 of 10 (Start time: 04:00:51 PM, 30 July 2018) Total Marks: 1

A sufficient condition for a function of several variables to be differentiable at point is

Select the correct option

Reload Math Equations

- ☐ only the partial derivative should exists at that point.
- ☒ All partial derivatives exists and are continuous at that point
- ☐ only limit at that point should exists.
- ☐ None of these

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Quiz Start Time: 03:53 PM, 30 July 2018

Question # 9 of 10 (Start time: 04:01:11 PM, 30 July 2018)

Total Marks: 1

along the line

$$y = -x$$

is

Select the correct option

Reload Math Equations

☐	$\frac{1}{2}$
☐	$-\frac{1}{2}$
☐	0
☐	1

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Quiz Start Time: 04:51 PM, 30 July 2018

Question # 1 of 10 (Start time: 04:51:01 PM, 30 July 2018)

Total Marks: 1

A point on the graph of a function at which the tangent line is parallel to x-axis is called the

Select the correct option

☒	critical point	//
☐	point of inflection	//
☐	point of relative extrema	//
☐	None of these	//

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Quiz Start Time: 04:51 PM, 30 July 2018

Question # 2 of 10 (Start time: 04:51:15 PM, 30 July 2018)

Total Marks: 1

A set $A \subset \mathbb{R}$ of real numbers is -----if there exists a real number $m \in \mathbb{R}$, such that $x \geq m$ for every $x \in A$.

Select the correct option

☐ bounded below

☐ uniformly continuous

☐ bounded above

☐ None of these

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Question # 3 of 10 (Start time: 04:52:52 PM, 30 July 2018)

Total Marks: 1

Identify the true statement(s)

Select the correct option

Reload Math Equations

☐

the function

$$f(x, y) = x^3 + y^3$$

has a local extreme point at origin.

☒

If

$$f_{x_i}(\mathbf{X}_0) = 0$$

then

☐

At every point where

$$f_{x_i}(\mathbf{X}_0) = 0$$

function has local extreme point.

☐

None of these



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Quiz Start Time: 04:51 PM, 30 July 2018

Question # 4 of 10 (Start time: 04:53:47 PM, 30 July 2018)

Total Marks: 1

, a point

$$\mathbf{x}_0$$

is local extreme point if

Select the correct option

Reload Math Equations



for all

$$\mathbf{x} \in D_f$$



partial derivatives exists and equal to zero.



does not change sign in

$$S_\delta(\mathbf{x}_0) \cap D_f$$



None of these

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Time Left 74 sec(s)

Quiz Start Time: 04:51 PM, 30 July 2018

Question # 5 of 10 (Start time: 04:54:32 PM, 30 July 2018)

Total Marks: 1

 $df =$

Select the correct option

Reload Math Equations

9

☐

9dx

☒

-1

☐

None of these

☐

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Time Left 27
sec(s)

Quiz Start Time: 04:51 PM, 30 July 2018

Question # 6 of 10 (Start time: 04:54:54 PM, 30 July 2018)

Total Marks: 1

For the function

$$f(x, y) = 3x^2y^3 + xy$$

in two variables,

Select the correct option

Reload Math Equations

☐	0
☐	-1
☒	1
☐	None of these



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Quiz Start Time: 04:51 PM, 30 July 2018

Question # 7 of 10 (Start time: 04:56:19 PM, 30 July 2018)

Total Marks: 1

and

$$f_{yx}(x_0, y_0)$$

exists. Then,

Select the correct option

Reload Math Equations

☒	$f_{yx}(x_0, y_0) = f_{xy}(x_0, y_0)$
☐	$f_{xx}(x_0, y_0) = f_{yy}(x_0, y_0)$
☐	$f_x(x_0, y_0) = f_y(x_0, y_0)$
☐	None of these

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Time Left 73
sec(s)

Quiz Start Time: 04:51 PM, 30 July 2018

Question # 8 of 10 (Start time: 04:56:40 PM, 30 July 2018)

Total Marks: 1

$$f(x, y)$$

as

$$(x, y) \rightarrow (0, 0)$$

Select the correct option

Reload Math Equations

☐

$$\frac{1}{2}$$

☒

$$-\frac{1}{2}$$

☐

0

☐

1

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Time Left 76 sec(s)

Quiz Start Time: 04:51 PM, 30 July 2018

Question # 10 of 10 (Start time: 04:57:08 PM, 30 July 2018)

Total Marks: 1

A tangent plane to the surface

$$z = f(x, y)$$

is always the limit of the -----

Select the correct option

Reload Math Equations

☒	secant plane
☐	tangent line
☐	normal plane
☐	None of these

Click to Save Answer & Move to Next Question

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Quiz

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Time Left 26 sec(s)

Quiz Start Time: 02:28 PM, 01 August 2018

Question # 4 of 10 (Start time: 02:29:13 PM, 01 August 2018)

Total Marks: 1

Identify the true statement(s)

Select the correct option

Reload Math Equations

☐ the function

$$f(x,y) = x^3 + y^3$$

has a local extreme point at origin.

☐ may or may not has a local extreme point at

$$X_0$$

☐ At every point where

$$f_{x_i}(X_0) = 0$$

function has local extreme point.

☐ None of these

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Time Left 57 sec(s)

Quiz Start Time: 03:53 PM, 30 July 2018

Question # 6 of 10 (Start time: 03:59:19 PM, 30 July 2018)

Total Marks: 1

$$f(x) = 3x^2 + 5x^3,$$

then at

$$\int_{-1}^1 f(x) dx =$$

Reload Math Equations

Select the correct option

- | | |
|----------------------------------|---------------|
| ☒ | 9 |
| ☐ | 9dx |
| ☐ | -1 |
| ☐ | None of these |

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Time Left 40 sec(s)

Quiz Start Time: 03:53 PM, 30 July 2018

Question # 10 of 10 (Start time: 04:02:09 PM, 30 July 2018) Total Marks: 1

, a point

$$\mathbf{x}_0$$

is local extreme point if

Select the correct option

Reload Math Equations

- ☐ for all
$$\mathbf{x} \in D_f$$
- ☐ partial derivatives exists and equal to zero.
- ☒ does not change sign in
$$S_\delta(\mathbf{x}_0) \cap D_f$$
- ☐ None of these

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Time Left 77 sec(s)

Quiz Start Time: 02:22 PM, 01 August 2018

Question # 8 of 10 (Start time: 02:23:58 PM, 01 August 2018) Total Marks: 1

For the functions $f(x) = 3x^2 + 5x^3$, the differential at $x = -1$ exists because

Select the correct option

Reload Math Equations

- ☐ $\lim_{x \rightarrow -1} \frac{f(x)}{x+1} = \infty.$
- ☐ $\lim_{x \rightarrow -1} \frac{9(x+1)}{x+1} = 0.$
- ☒ $\lim_{x \rightarrow -1} \frac{f(x) + 2 - 9(x+1)}{x+1} = 0.$
- ☐ $\lim_{x \rightarrow -1} \frac{f(x) + 2 - 9(x+1)}{x+1} = 1.$

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Example: Determine the convergence of the improper integral

$$I = \int_0^1 \frac{2 + \sin \pi x}{(1 - x)^p} dx.$$

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Time Left 66 sec(s)

Quiz Start Time: 02:22 PM, 01 August 2018

Question # 9 of 10 (Start time: 02:24:15 PM, 01 August 2018)

Total Marks: 1

A function may be at a point

X_0

even if its first partial derivatives are not continuous at

Select the correct option

Reload Math Equations

- ☐ uniformly convergent
- ☒ differentiable
- ☐ piecewise convergent
- ☐ None of these

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Time Left 25 sec(s)

Quiz Start Time: 02:22 PM, 01 August 2018

Question # 10 of 10 (Start time: 02:24:46 PM, 01 August 2018) Total Marks: 1

Identify the true statement(s)

Select the correct option

Reload Math Equations

- ☐ A function of several variables is differentiable if all of its partial derivatives exists.
- ☐ A function of several variables is continuous then it is differentiable.
- ☐ $L(x) = 2x - 3x^2$ is a linear function.
- ☒ is $\frac{\partial f(\mathbf{X})}{\partial \mathbf{E}_2}$

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f is continuously differentiable on S .

Remark: If $f_{x_1}, f_{x_2}, \dots, f_{x_n}$ exist on a neighborhood of $\mathbf{X}_0$ and are continuous at $\mathbf{X}_0$, then f is differentiable at $\mathbf{X}_0$. These conditions are not necessary for differentiability; that is, a function may be differentiable at a point $\mathbf{X}_0$ even if its first partial derivatives are not continuous at $\mathbf{X}_0$.

Example: let

$$f(x, y) = \begin{cases} (x - y)^2 \sin \frac{1}{x - y}, & x \neq y, \\ 0, & x = y. \end{cases}$$

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Then

$$f_x(x, y) = 2(x - y) \sin \frac{1}{x - y} - \cos \frac{1}{x - y}, \quad x \neq y,$$

and

$$f_x(x, x) = \lim_{h \rightarrow 0} \frac{f(x + h, x) - f(x, x)}{h} = \lim_{h \rightarrow 0} \frac{h^2 \sin(1/h) - 0}{h} = 0,$$

so f_x exists for all (x, y) , but is not continuous on the line $y = x$.

Example: Let

$$f(x, y) = \begin{cases} (x - y)^2 \sin \frac{1}{x - y}, & x \neq y, \\ 0, & x = y. \end{cases}$$

The same is true of f_y , since

$$f_y(x, y) = -2(x - y) \sin \frac{1}{x - y} + \cos \frac{1}{x - y}, \quad x \neq y,$$

and

$$f_y(x, x) = \lim_{k \rightarrow 0} \frac{f(x, x + k) - f(x, x)}{k} = \lim_{k \rightarrow 0} \frac{k^2 \sin(-1/k) - 0}{k} = 0.$$

Now,

$$\begin{aligned} & \frac{f(x, y) - f(0, 0) - f_x(0, 0)x - f_y(0, 0)y}{\sqrt{x^2 + y^2}} \\ &= \begin{cases} \frac{(x - y)^2}{\sqrt{x^2 + y^2}} \sin \frac{1}{x - y}, & x \neq y, \\ 0, & x = y, \end{cases} \end{aligned}$$

and Schwarz's inequality implies that

$$\left| \frac{(x - y)^2}{\sqrt{x^2 + y^2}} \sin \frac{1}{x - y} \right| \leq \frac{2(x^2 + y^2)}{\sqrt{x^2 + y^2}} = 2\sqrt{x^2 + y^2}, \quad x \neq y.$$

Therefore,

$$\lim_{(x, y) \rightarrow (0, 0)} \frac{f(x, y) - f(0, 0) - f_x(0, 0)x - f_y(0, 0)y}{\sqrt{x^2 + y^2}} = 0,$$

so f is differentiable at $(0, 0)$, but f_x and f_y are not continuous at $(0, 0)$.

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- ⑤ $f(x,y) = \frac{x^2+y^2}{x-y}$ → Show That f is differentiable on S .
(2,2) → ⑤
- ④ Suppose $U_0 \in T$ is a limit point of T , g is continuous at $x_0 = g(U_0)$.
Then show that $f \circ g$ is continuous at U_0 . → ⑩
- ① S_1 has zero content and $S_2 \subset S_1$, then S_2 has zero content. → ⑤
- ② Find the value $\int_0^\infty x^{-p} dx$ exists an improper integral → ⑩
- ③ $P(x,y) = x^2 - 2xy + y^2 - x^4 - y^4$ → ⑤
Show That $(0,0)$ is a local point
and it has positive semidefinite
second differentiable at $(0,0)$

Qn.1 Find upper and lower integrals in $\mathbb{R}^n$

For a partition $P = \{R_1, R_2, R_3, \dots, R_k\}$ of R . (10)

Qn.2

Evaluate

limit where $f(x,y) = \frac{\sin(x-y)}{\sqrt{x-y}}$ (10)
 $x, y \rightarrow (2,2)$

Qn.3

Define Absolute Integral

Dirichlet's test (5)

Qn.4

Determine Integral

$$\int_1^{\infty} \frac{\log x + \sin x}{\sqrt{x}} \text{ Converges or Diverges.}$$

Qn.5

Calculate

$$\frac{\partial f(x)}{\partial \phi}$$

where $f(x,y,z) = \log(1+x+y+z)$

$$\phi(0,1,0) \quad (5 \text{ marks})$$

Qn.6

Define ~~Diameter~~ Diameter of a Set

$$\text{if } S = \{(x,y,z), |x| \leq 2, |y| \leq 1, |z-2| \leq 2\}$$

Find $d(S)$

Q:- Show that interval is the only region of R .

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2.17. Differentiability of Functions of Several Variables

2.17.5 Geometric Interpretation of Differentiability

If a function f of one variable is differentiable at x_0 , then the curve $y = f(x)$ has a unique tangent line

$$y = T(x) = f(x_0) + f'(x_0)(x - x_0).$$

The tangent line approximates it so well near x_0 that

$$\lim_{x \rightarrow x_0} \frac{f(x) - T(x)}{x - x_0} = 0.$$

Moreover, the tangent line is the limit of the secant line through $(x_0, f(x_0))$ and $(x_1, f(x_1))$ as x_1 approaches x_0 . Differentiability of

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If $\lim_{|(x,y,z)| \rightarrow \infty} f(x,y,z) = 1$ then prove that $f(x) = \cos\left(\frac{1}{x+2y+z}\right)$ is differential.

2.17. Differentiability of Functions of Several Variables

105

2.17.2 Differential

The linear function

$$L(\mathbf{X}) = f_{x_1}(\mathbf{X}_0)x_1 + f_{x_2}(\mathbf{X}_0)x_2 + \cdots + f_{x_n}(\mathbf{X}_0)x_n.$$

This function is called the *differential of f at $\mathbf{X}_0$* . We will denote it by $d_{\mathbf{X}_0}f$ and its value by $(d_{\mathbf{X}_0}f)(\mathbf{X})$.

Thus,

$$(d_{\mathbf{X}_0}f)(\mathbf{X}) = f_{x_1}(\mathbf{X}_0)x_1 + f_{x_2}(\mathbf{X}_0)x_2 + \cdots + f_{x_n}(\mathbf{X}_0)x_n. \quad (2.50)$$

... polar coordinates in the xy -plane; that is,
 $x = r \cos \theta, \quad y = r \sin \theta.$

that $f = f(x, y)$ is differentiable on a set S , and let

$$h(r, \theta) = f(r \cos \theta, r \sin \theta).$$

most important MCQ

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$$\frac{\partial h}{\partial r} = \frac{\partial f}{\partial x} \frac{\partial x}{\partial r} + \frac{\partial f}{\partial y} \frac{\partial y}{\partial r} = \cos \theta \frac{\partial f}{\partial x} + \sin \theta \frac{\partial f}{\partial y} \quad (2.85)$$

$$\left[\frac{\partial h}{\partial \theta} = \frac{\partial f}{\partial x} \frac{\partial x}{\partial \theta} + \frac{\partial f}{\partial y} \frac{\partial y}{\partial \theta} = -r \sin \theta \frac{\partial f}{\partial x} + r \cos \theta \frac{\partial f}{\partial y} \right] \leftarrow \text{mcq's}$$

and f_y are evaluated at $(x, y) = (r \cos \theta, r \sin \theta).$

e: Suppose that f_x and f_y just calculated are differentiable on an open set
 Differentiating (2.85) with respect to r yields

$$\begin{aligned} \frac{\partial^2 h}{\partial r^2} &= \cos \theta \frac{\partial}{\partial r} \left(\frac{\partial f}{\partial x} \right) + \sin \theta \frac{\partial}{\partial r} \left(\frac{\partial f}{\partial y} \right) \\ &= \cos \theta \left(\frac{\partial^2 f}{\partial x^2} \frac{\partial x}{\partial r} + \frac{\partial^2 f}{\partial y \partial x} \frac{\partial y}{\partial r} \right) + \sin \theta \left(\frac{\partial^2 f}{\partial x \partial y} \frac{\partial x}{\partial r} + \frac{\partial^2 f}{\partial y^2} \frac{\partial y}{\partial r} \right). \end{aligned} \quad (2.86)$$

Definition ?? yields

$$\int_0^1 \log x \, dx = -1.$$

The function $f(x) = \cos x$ is locally integrable on $[0, \infty)$ and

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$$\lim_{c \rightarrow \infty} \int_0^c \cos x \, dx = \lim_{c \rightarrow \infty} \sin c$$

fine

does not exist; thus, $\int_0^\infty \cos x \, dx$ diverges, but not to $\pm\infty$.

In connection with Definition ??, it is important to recognize that the improper integrals $\int_a^\alpha f(x) \, dx$ and $\int_\alpha^b f(x) \, dx$ must converge *separately* for $\int_a^b f(x) \, dx$ to converge. For example, the existence of the symmetric limit

$$\lim_{R \rightarrow \infty} \int_{-R}^R f(x) \, dx,$$

which is called the *principal value* of $\int_{-\infty}^\infty f(x) \, dx$, does not imply that $\int_{-\infty}^\infty f(x) \, dx$ converges; thus,

$$\lim_{R \rightarrow \infty} \int_{-R}^R x \, dx = \lim_{R \rightarrow \infty} 0 = 0,$$

Q2- S_1 & S_2 has zero content then prove that $S_1 \cup S_2$ has zero content.

3.12. Differentiable Surfaces

155

Theorem: If f and g are integrable on S and $f(\mathbf{X}) \leq g(\mathbf{X})$ for $\mathbf{X}$ in S , then

$$\int_S f(\mathbf{X}) d\mathbf{X} \leq \int_S g(\mathbf{X}) d\mathbf{X}.$$

mcq/s

Theorem: If f is integrable on S , then so is $|f|$, and

$$\left| \int_S f(\mathbf{X}) d\mathbf{X} \right| \leq \int_S |f(\mathbf{X})| d\mathbf{X}.$$

Theorem: If f and g are integrable on S , then so is the product fg .

Theorem: Suppose that u is continuous and v is integrable and nonnegative on a

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Example
zero content.

If E consists of finitely many points $X_1, X_2, \dots, X_m$, then X_j can be enclosed in a rectangle T_j such that

$$V(T_j) < \frac{\varepsilon}{m}, \quad 1 \leq j \leq m.$$

Then $E \subset \bigcup_{j=1}^m T_j$ and $\sum_{j=1}^m V(T_j) < \varepsilon$ hold, so E has zero content.

Example: Any bounded set E with only finitely many limit points has zero content. *mcg's*

To see this, we first observe that if E has no limit points, then it must be finite, by the Bolzano-Weierstrass theorem, and therefore must have zero content.

Now suppose that the limit points of E are $X_1, X_2, \dots, X_m$. Let $R_1, R_2, \dots, R_m$ be rectangles such that $X_i \in R_i^0$ and

$$V(R_i) < \frac{\varepsilon}{2m}, \quad 1 \leq i \leq m. \quad (3.30)$$

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The set of points of E that are not in $\bigcup_{j=1}^m R_j$ has no limit points (why?) and, being bounded, must be finite (again by the Bolzano-Weierstrass theorem).

If this set contains p points, then it can be covered by rectangles $R'_1, R'_2, \dots, R'_p$ with

$$V(R'_j) < \frac{\varepsilon}{2p}, \quad 1 \leq j \leq p. \quad (3.31)$$

Now,

$$E \subset \left(\bigcup_{i=1}^m R_i \right) \cup \left(\bigcup_{j=1}^p R'_j \right)$$

(3.22) implies (3.21). $|\sigma - \sigma'| = |f(\mathbf{X}) - f(\mathbf{X}_i)|V(R_i),$

3.9 Upper and Lower Integrals

If f is bounded on a rectangle R in $\mathbb{R}^n$ and $\mathbf{P} = \{R_1, R_2, \dots, R_k\}$ is a partition of R .

Let

$$M_j = \sup_{\mathbf{X} \in R_j} f(\mathbf{X}), \quad m_j = \inf_{\mathbf{X} \in R_j} f(\mathbf{X}).$$

The upper sum of f over $\mathbf{P}$ is

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$$S(\mathbf{P}) = \sum_{j=1}^k M_j V(R_j).$$

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